Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 23-03-2026

PROJECT TITLE:Academic Campus Event & Registration Portal

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

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INTRODUCTION

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- Campus events often rely on fragmented communication like posters and manual registration processes.
- Centralizing this information improves student engagement and administrative efficiency.
- Universities need modern, accessible platforms to manage extracurriculars
- This project is a responsive, 3-tier web application that allows students to securely discover and register for campus events.

PROBLEM STATEMENT

- Currently, academic and cultural events are managed through disjointed channels, leading to low visibility.
- The manual registration process causes inefficiency, potential data loss, and a poor user experience.
- Organizers struggle to track participant data and manage event capacities effectively.
- There is a clear need for a centralized, automated web portal to streamline event discovery and secure student registrations in real-time.

OBJECTIVE(S)

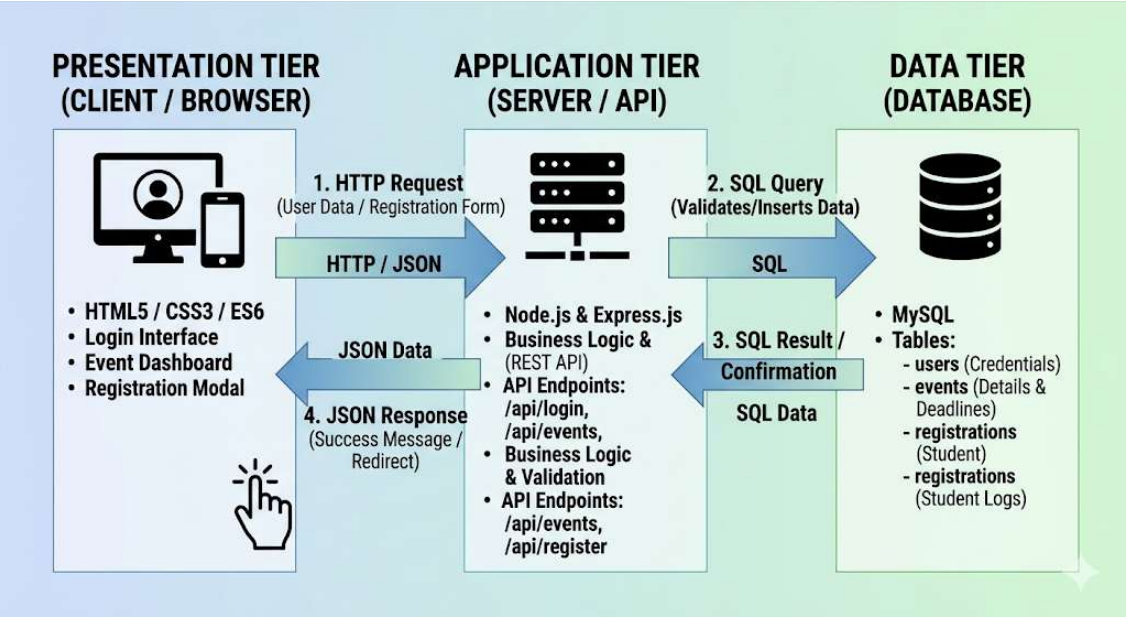
- To design and develop a dynamic, full-stack web application that centralizes campus event discovery and categorization.

- To implement a secure, automated registration system that captures student data seamlessly using a MySQL backend and Node.js REST API.

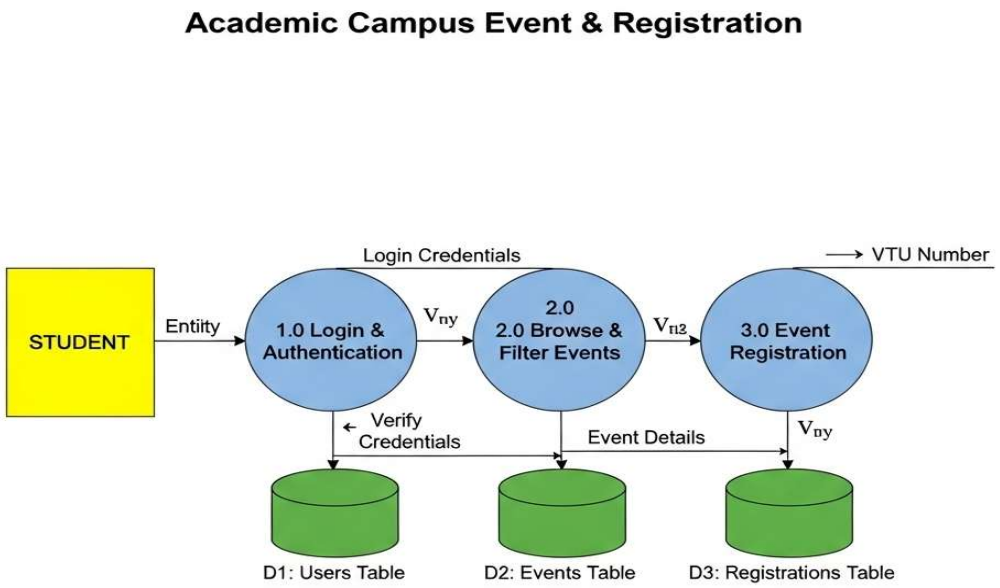
PROPOSED SYSTEM

- A complete 3-tier architecture application (Frontend, Backend, Database).
- User authentication system for secure student login and sign-up.
- Interactive dashboard with real-time search and category filtering (Technical, Cultural, Seminar).
- Dynamic data retrieval from a centralized MySQL database.
- Automated registration form with auto-filled user details via session storage.
- Eliminates manual paperwork and fragmented communication.
- Provides an intuitive interface using modern HTML5, CSS3, and ES6.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW


Academic Campus Event & Registration
STUDENT
Entity
1.0 Login & Authentication
Login Credentials
2.0
2.0 Browse & Filter Events
3.0 Event Registration
VTU Number
Verify Credentials
Event Details
D1: Users Table
D2: Events Table
D3: Registrations Table

TECHNOLOGIES USED

- **Frontend:** HTML5, CSS3, JavaScript (React or Bootstrap for styling).
- **Backend:** Node.js, Express.js (REST APIs)
- **Database:** MySQL
- **Tools:** VS Code, MySQL Workbench
- **Hardware/Software Requirements:** Windows/macOS, Modern Web Browser, Node Runtime Environment.

MODULE DESCRIPTION

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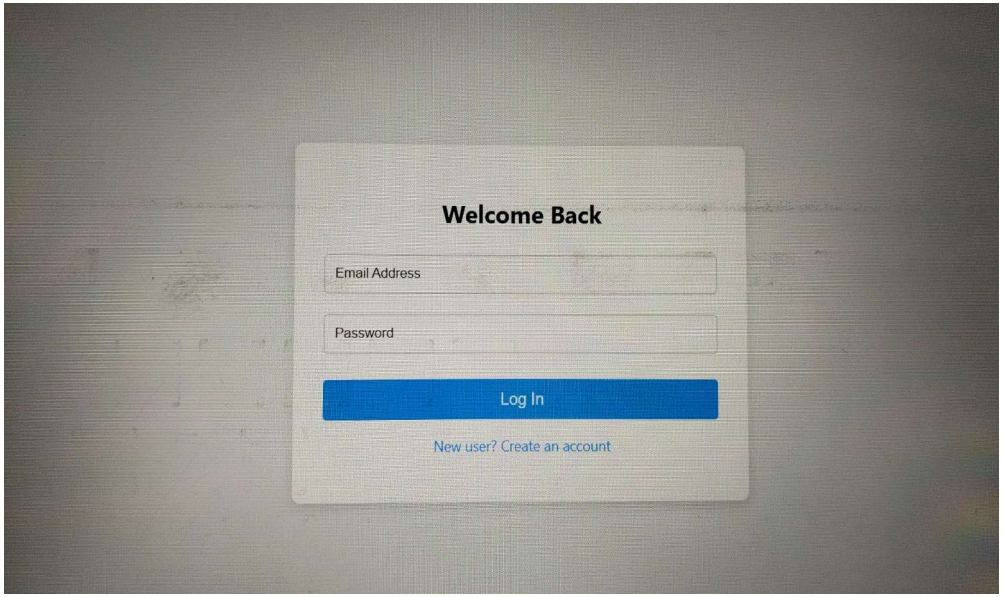
----- Example for reference only ---- (Change according to your project)

Module 1: User Management & Authentication
Handles student sign-up, secure login verification, and browser session state management.
Module 2: Dynamic Event Dashboard
. Retrieves event data, renders dynamic UI cards, and processes live search and category filtering.
Module 3: Registration Engine
Manages the modal popup, captures VTU credentials, and successfully posts data to the backend API.
Module 4: Database Management

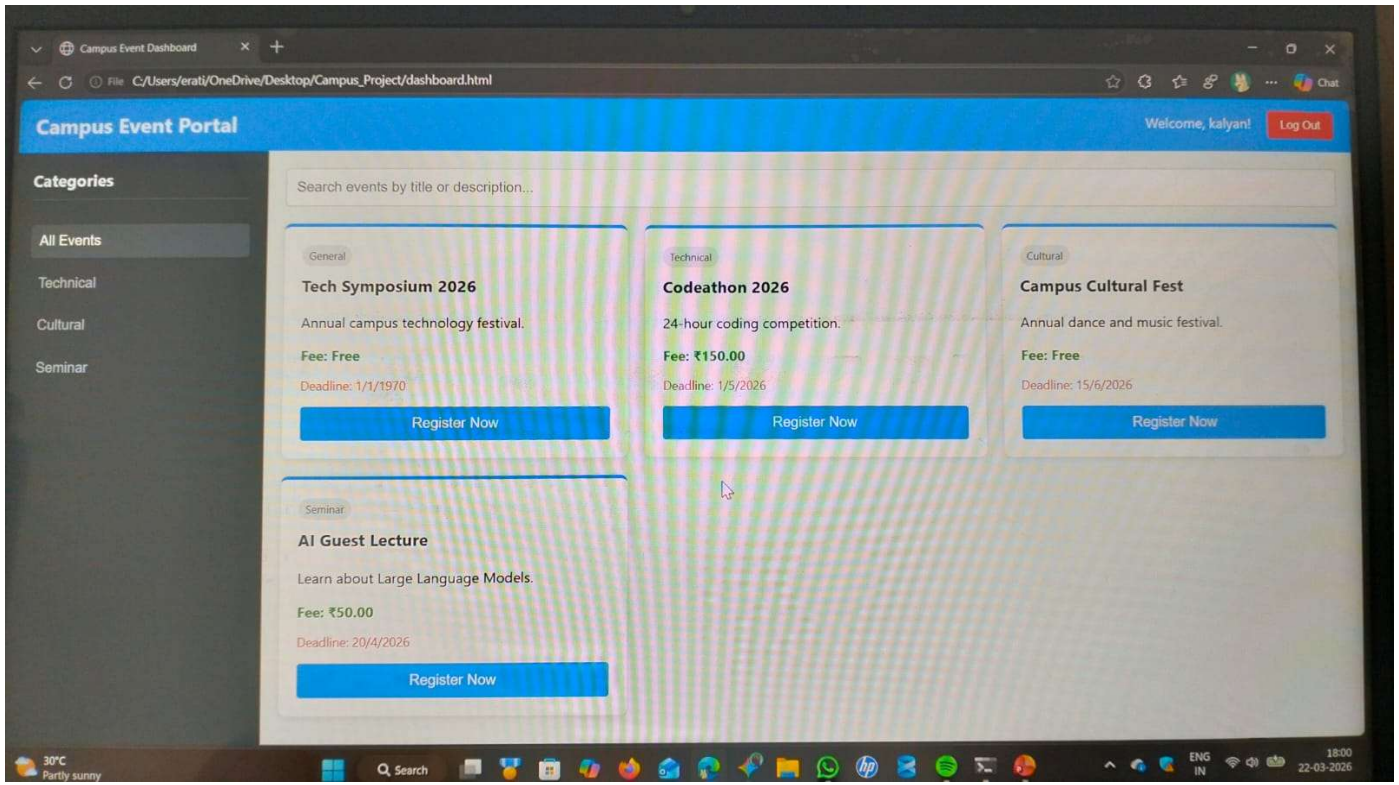
IMPLEMENTATION SCREENSHOTS



- **Login page**

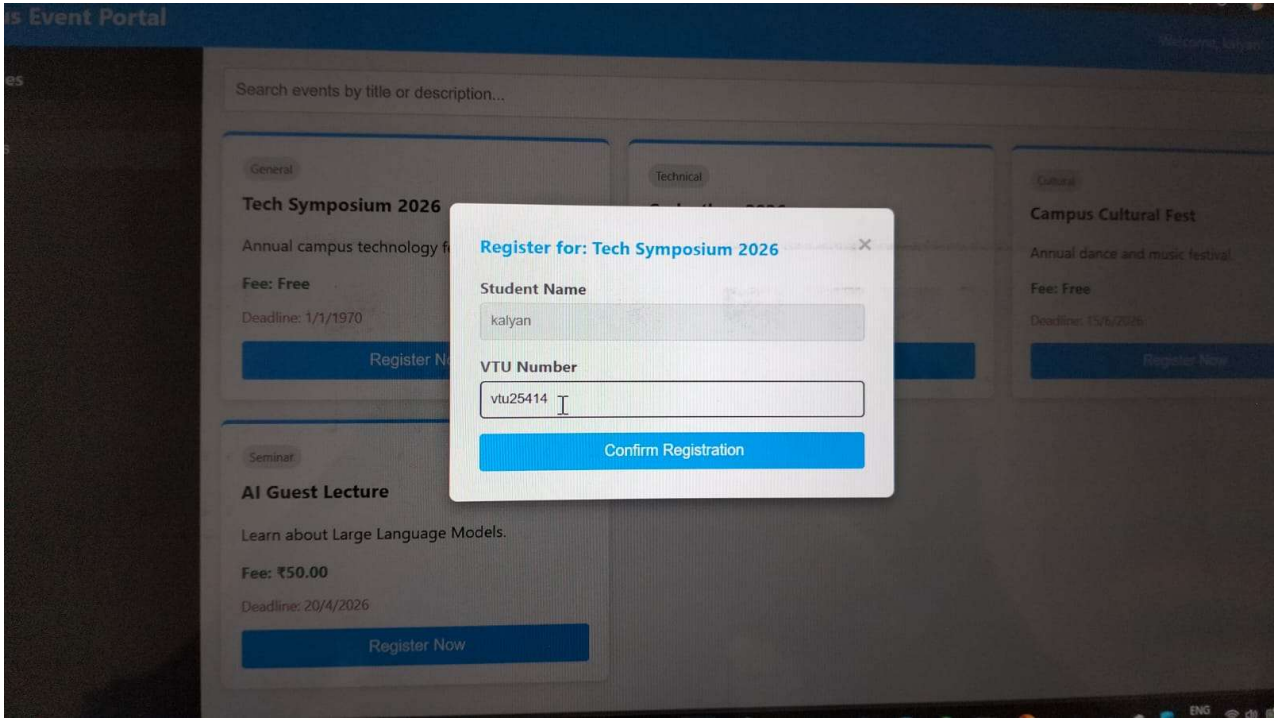
- Secure user authentication and account creation interface

IMPLEMENTATION SCREENSHOTS



- **Main Dashboard**
- The dynamic event dashboard featuring category filtering and search capabilities.

IMPLEMENTATION SCREENSHOTS



- **Registration Modal**

- Automated registration modal that auto-fills the logged-in student's details.

IMPLEMENTATION SCREENSHOTS

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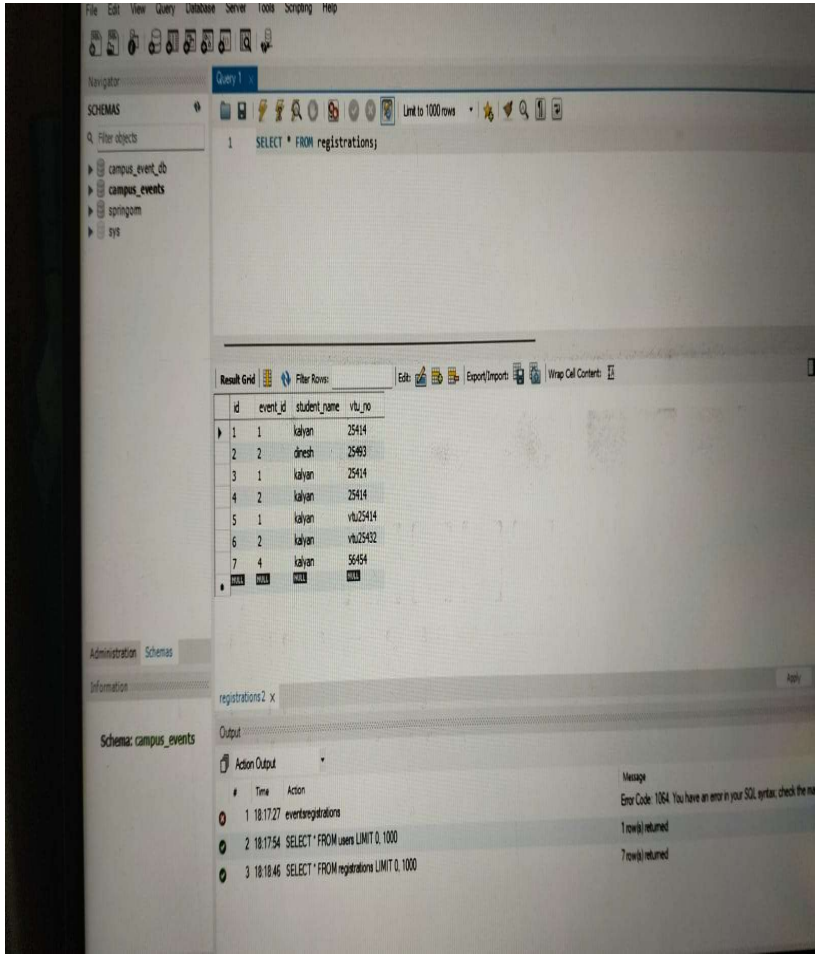

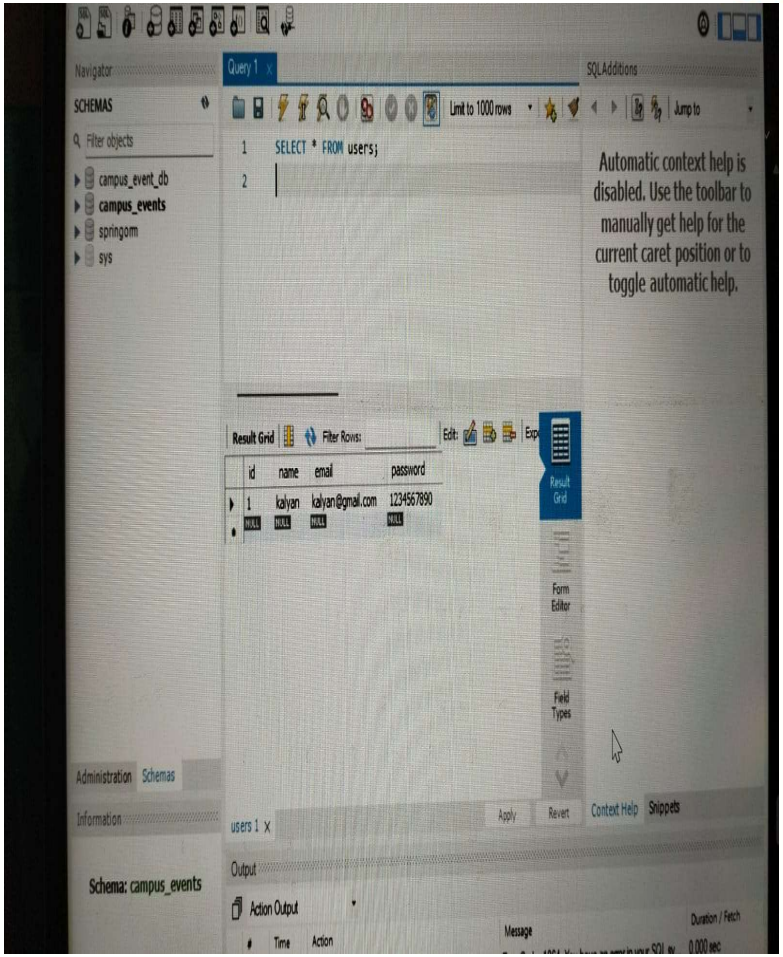

Output/Database Page

ADVANTAGES & LIMITATIONS

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Advantages:
Real-time event updates and paperless registration.
Centralized data access for both students and organizers..
Highly scalable architecture separating the frontend UI from backend data
.
Limitations:
Requires a constant internet connection to function.
Lacks an integrated payment gateway for events that require a fee.
Passwords are currently stored in plain text (can be upgraded with hashing in future iterations).

Thank you